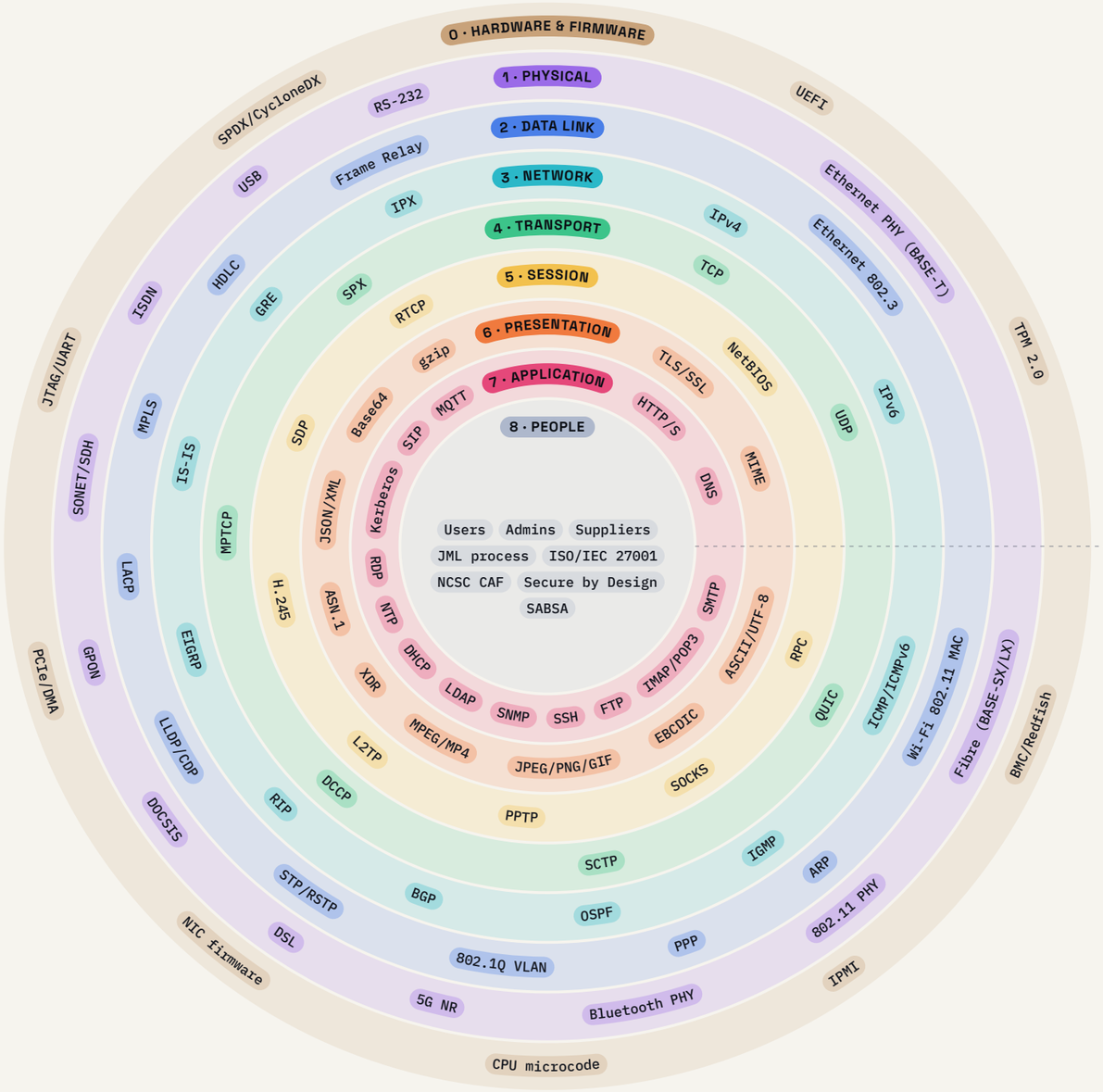
OSI Model — Protocols & Security Technologies by Layer

The seven-layer OSI model (ISO/IEC 7498-1) extended with Layer 0 (hardware, firmware & supply chain) on the outer ring and Layer 8 (people & governance) at the core.

SECURITY ARCHITECTURE REFERENCE
DEFENCE IN DEPTH · LAYER BY LAYER
OSI.CYBERASSURE.UK · v2.0 · OCTOBER 2026

RING SET A · SENDING STACK · WHAT TRAVERSES EACH LAYER

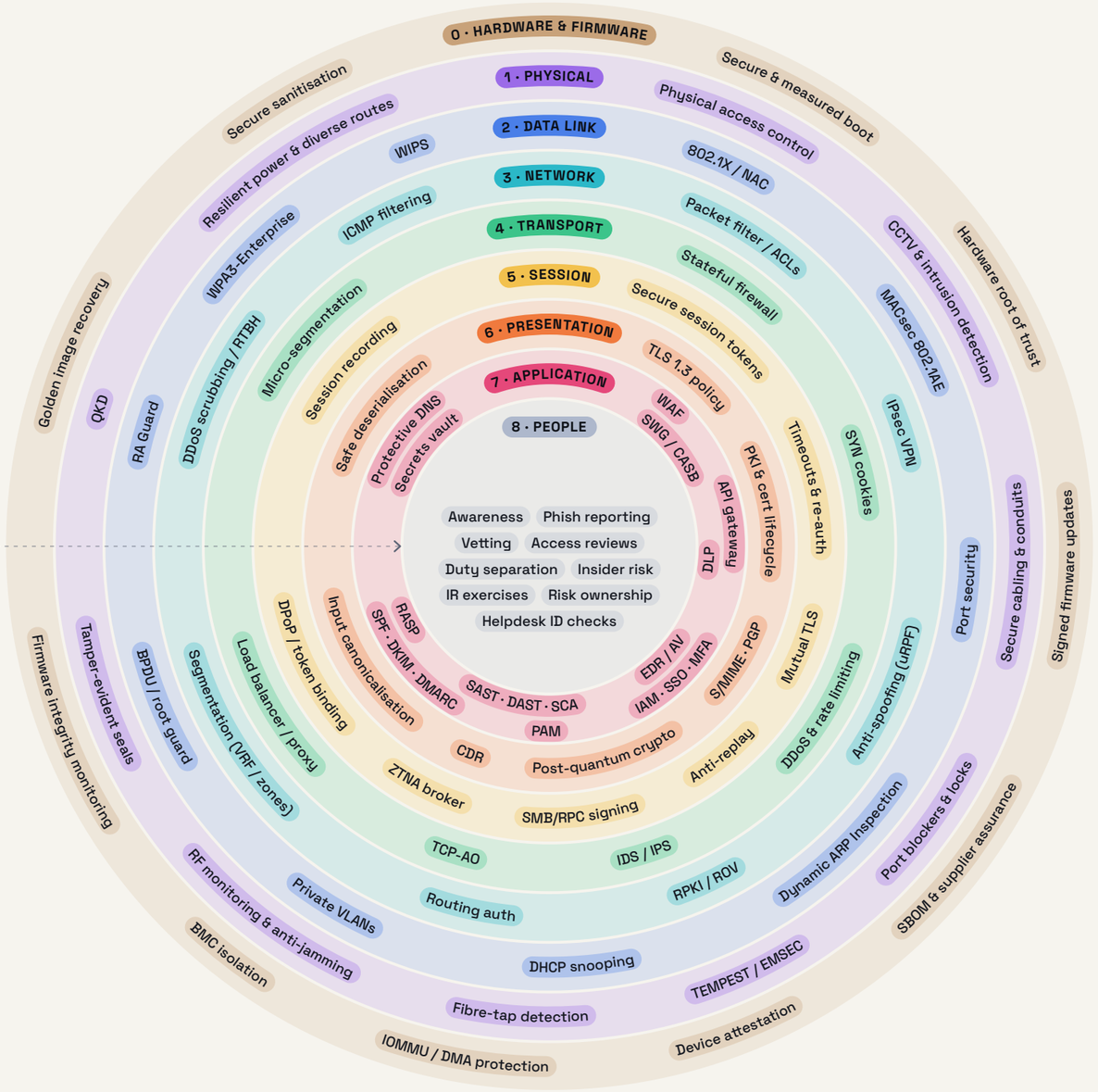
Protocols & Standards



WIRE
>>
BITS

RING SET B · RECEIVING STACK · HOW EACH LAYER IS DEFENDED

Security Technologies & Controls



LAYER KEY · DATA UNIT · CONTROLS BY FUNCTION (PREVENT · DETECT · RESPOND)

0 **Hardware** Silicon & firmware P7 · D3 · R1	1 **Physical** Bits / Symbols P5 · D4 · R1	2 **Data Link** Frame P9 · D1 · R1	3 **Network** Packet P7 · D0 · R2	4 **Transport** Segment / Datagram P7 · D1 · R2	5 **Session** Data P7 · D1 · R1	6 **Presentation** Data P7 · D0 · R0	7 **Application** Data P11 · D3 · R1	8 **People** Human intent P6 · D3 · R1

CROSS-LAYER CAPABILITIES

Zero Trust architecture · SIEM / SOAR · NDR · XDR · NGFW (L3–L7) · Threat intelligence · Asset & config management · Vulnerability management

Mapping is indicative: several protocols span layers or sit differently in the TCP/IP model (e.g. TLS spans L4–L7; BGP runs over TCP but is charted at L3 for its routing role).
Teaching model: ring position does not imply trust — apply Zero Trust at every ring. Controls sit at the layer they primarily act on; P / D / R = prevent, detect, respond.

Jon Silvester · linkedin.com/in/jonsilvester